

Questions, and where the data comes from

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Research Design for Social Sciences
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Where we are

1. Why design at all? done
2. **What is the question, and where would the data come from? today**
3. Do the numbers mean what you think they mean?
4. What comparisons can we use to establish causal claims?
5. What variation can you exploit, and how far does the answer travel?

Roadmap

From topic to research question

Theory, mechanisms, levels

Where does the data come from?

Ten things about big data sources

So where do you actually get data?

Paper: The Parable of Google Flu

Before next week

Research and RQs

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- Re-cap on research types:
 1. **Normative** vs. **positive** research

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 2. Positive: **theoretical** vs. **empirical**

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- Re-cap on research types:
 1. **Normative** vs. **positive** research
 2. Positive: **theoretical** vs. **empirical**
 3. Empirical: **describe**, **explain**, **predict**

Example on types of questions and evidence

- You are doing the analytics behind a campaign for Mahou
- Think of questions that are:
 - Descriptive
 - Explanatory
 - Predictive
- Which ones could you answer with the data a brewer actually has?



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- **Even though** this 'argument' can be something anecdotal that we later develop into a proper, abstract theoretical argument
 - And it would actually look into something like this:
 - Previous > 'Anecdotal argument' > RQ > (Proper) Theory > Hs

What is a research question? Example

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- 'Main' RQ: Does online media cause an increase in lone-wolf political violence?
 - RQ1: Has political violence increased recently?
 - RQ2: Political violence now more likely by individual young men?
 - RQ3: Is online radicalization the cause of this?

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 - RQ3: Is online radicalization the cause of this?
- Theoretical argument, hypotheses, expectations, etc

Good RQs, in brief

1. Empirically **answerable**

→ i.e. you can answer it with data

2. Theoretically **relevant**

→ i.e. it helps you learn something about your theory/argument

Good RQs, more in detail

1. Consider potential results of the analyses

- if you found X, does that answer the question? causally?
- example: are kids who play videogames often more aggressive?
- does that inform a theory on the aggressiveness effect of VGs?
(you can even try to do a better RQ without causal ID)

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2. Is it feasible?

- do you have the data? is it possible to do it? (e.g. re-offenders)
- also: is there any design or strategy to answer it?

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3. Keeping it simple and narrow

- what are the causes of economic underdevelopment? vs. does exposure to natural disasters hinders economic development?

Good RQs, more in detail

This:

- Has political violence increased recently?
- Political violence now more likely by individual young men?
- Is online radicalization the cause of this?

vs this:

- What's the link between online media and political violence?
(too broad)
- Is online media the cause of the increase of political violence among young men? (many questions in one)
- Why does online media drive political violence? (??)

Example on generating RQs

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 - RQs are often the link between theory and empirics
 - So they already suggest which variation to look at
- Imagine you have the following argument:

How good students do at school depends more on the peers they are surrounded by than on the quality of the teaching they receive
- Which RQ could let us test this?

Stories, RQs, and theories

- There're no exclusive definitions of 'stories' and theories
- It's just about getting to a sufficient level of abstraction
- Often, you start with a story or example, and then you move up the ladder for both theory and RQs until you get to a general theory tested with a RQ

Generating theories

- No recipe for this, everyone generates theories *all the time*
- Usually it refers to an analytical argument that explains something
 - It could also be a descriptive or predictive theory, but even in those cases there's probably an explanation underneath
- Developed inductively, from descriptive data to general explanations
- My advice: if you can't tell a story out of the theory, you're not there yet (i.e. need to be able to travel from/to abstraction)

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- **Q:** How to identify a **good theory**?

Evaluating theories

1. Simple

Evaluating theories

1. Simple
2. Internally coherent and able to explain variation

Evaluating theories

1. Simple
2. Internally coherent and able to explain variation
3. Testable

Example (of the whole process)

Internacionales UMU
@internacionalUMU

¿Y sabías que encuentran **#trabajo** más rápido que los estudiantes que no realizan **#movilidad**?

3 de cada 4 graduados Erasmus+ consideran que su **#experiencia** en el extranjero es beneficiosa para encontrar su primer trabajo, y el 80% encontraron trabajo 3 meses después de graduarse.

Translate Tweet



The advertisement graphic features a blue background with a hand pointing at a glowing orange figure icon. To the right, there are two smaller blue circles, each containing a white figure icon. The text is in white and yellow. The top left corner has the Erasmus+ logo and the text 'Erasmus+ ESTUDIOS'. The main headline is 'DIFERENCIATE DEL RESTO'. Below it, a paragraph states: 'Los estudiantes Erasmus+ encuentran trabajo más rápido que los estudiantes que no realizan movilidad.' At the bottom left, vertical text reads 'ELIGE TU DESTINO ELIGE ERASMUS'. At the bottom center, it says 'Solicitud abierta hasta el 14/12/20' and 'erasmus.um.es'. At the bottom right, there is a red box with 'UNIVERSIDAD DE MURCIA' and a small circular logo.

ELIGE TU DESTINO
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UNIVERSIDAD DE
MURCIA

12:05 PM · Nov 23, 2020 · Twitter Web App

Example

- That's some descriptive evidence that could inspire an anecdote
- The **anecdotal argument** (think of a story)
- The **research question**
- The 'proper' **theory**
- The **hypotheses?**

Example

- That's some descriptive evidence that could inspire an anecdote
- The **anecdotal argument** (think of a story)
 - My friend John who went on Erasmus has more money than my other friend who couldn't go, and also, John managed to get a job because his father is partner at a local firm
- The **research question**
 - Is there a causal effect of Erasmus on labor market early success? Is the effect mediated by household income?
- The 'proper' **theory**
 - Going on Erasmus does not have any causal effect on getting a first job, the relationship is explained by the confounding effect of income
 - Or: Positive effect among high-income students because they have access to informal networks where this experience is valued
- The **hypotheses?**

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And ideally, knowing what you need to see in order to discredit the theory
- Imagine that I have the theory that my knee hurts when I do sports on cold days
 - Simplifying it, we have a 2x2 situation:

Hypotheses: what would you expect to observe?

	<i>Cold day</i>	<i>Hot day</i>
<i>Run</i>		
<i>Didn't run</i>		

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	<i>Cold day</i>	<i>Hot day</i>
<i>Run</i>	Pain	Not pain
<i>Didn't run</i>	Not pain	Not pain

What if you observe this? **New theory? Test?**

	<i>Cold day</i>	<i>Hot day</i>
<i>Run</i>	Pain	Not pain
<i>Didn't run</i>	Pain	Pain

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Before next week

Computational methods and theory

- Limitations of *data mining*
- Focus on the what rather than on the why
- Problems with machine learning
 - Example of predicting ice cream sales

Why bother about theory?

CIVIL UNREST

Predicting Civil Conflict: What Machine Learning Can Tell Us

Computer programs can be used as early warning systems, allowing the global community to act before violence erupts.

Mechanisms briefly

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- e.g., we know that the flu gives us fever
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Mechanisms briefly

- A **mechanism** is basically the **how** (or why) of a relationship
- e.g., we know that the flu gives us fever
 - flu > fever
- What's the mechanism?
 - flu > *immune system detects infection* > Δ *body temp* > fever
- Good thing about mechanisms is that we can try to test them

Testing mechanisms ($\approx$ sub-research questions)

- Let's go back to the Erasmus example
- If our theory is that the effect of going on Erasmus is higher for high-income students due to their access to social networks,
what's the mechanism?
- And how could we try to **test it**?
 - (Think about the sub-RQs)

Complexity of the social world and micro/macro



- Why do mass protests emerge?

Levels of explanation

Can you think of...?

- Macro-level mechanisms
- Micro-level mechanisms

Levels of explanation

Can you think of...?

- Macro-level mechanisms
- Micro-level mechanisms
- What's the point of macro-level explanations, actually?

Recap

- Questions on theory, RQ, or mechanisms?

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Last week, in one line

Designed

answers a question **you** had

Found

answers a question **someone else** had.
Or no question at all

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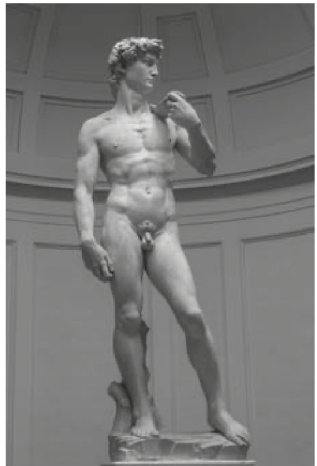
Found

answers a question **someone else** had.
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Today: what follows from being on the right-hand side.



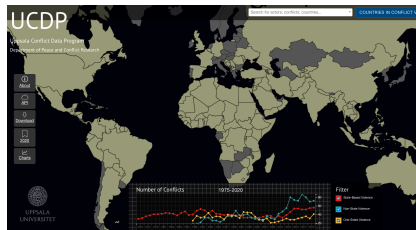
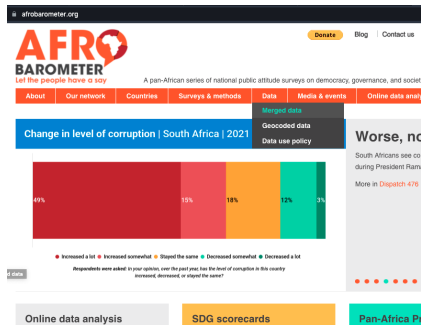
Readymade



Custommade

Figure 1.2: *Fountain* by Marcel Duchamp and *David* by Michelangelo. *Fountain* is an example of a readymade, where an artist sees something that already exists in the world and then creatively repurposes it for art. *David* is an example of art that was intentionally created; it is a custommade. Social research in the digital age will involve both readymades and custommades. Photograph of *Fountain* by Alfred Stieglitz, 1917 (Source: *The Blind Man*, no.

Designed data: someone decided what to record



A survey, and a hand-coded conflict dataset. Someone wrote the questions, or the coding rules.

Found data: nobody decided anything, for you

[nature](#) > [nature communications](#) > [articles](#) > [article](#)

Article | [Open Access](#) | [Published: 20 August 2018](#)

Sequences of purchases in credit card data reveal lifestyles in urban populations

[Riccardo Di Clemente](#), [Miguel Luengo-Oroz](#), [Matias Travizano](#), [Sharon Xu](#), [Bapu Vaitla](#) & [Marta C. González](#) 

[Nature Communications](#) **9**, Article number: 3330 (2018) | [Cite this article](#)

14k Accesses | 31 Citations | 268 Altmetric | [Metrics](#)

 This article has been [updated](#)

Abstract

Zipf-like distributions characterize a wide set of phenomena in physics, biology, economics, and social sciences. In human activities, Zipf's law describes, for example, the frequency of appearance of words in a text or the purchase types in shopping patterns. In the latter, the uneven distribution of transaction types is bound with the temporal sequences of purchases of individual choices. In this work, we define a framework using a text compression technique

RESEARCH ARTICLE

Faces in the crowd: Twitter as alternative to protest surveys

[Christopher Barrie](#) ^{1,*,} [Arun Frey](#) ^{2,3}

1 School of Social & Political Science, University of Edinburgh, Scotland, United Kingdom, **2** Department of Sociology, University of Oxford, England, United Kingdom, **3** Leverhulme Centre for Demographic Science, Oxford, England, United Kingdom

 All these authors contributed equally to this work and are listed in alphabetical order.

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Abstract

Who goes to protests? To answer this question, existing research has relied either on retrospective surveys of populations or in-protest surveys of participants. Both techniques are prohibitively costly and face logistical and methodological constraints. In this article, we investigate the possibility of surveying protests using Twitter. We propose two techniques for sampling protestors on the ground from digital traces and estimate the demographic and ideological composition of ten protestor crowds using multidimensional scaling and machine-learning techniques. We test the accuracy of our estimates by comparing to two in-protest surveys from the 2017 Women's March in Washington, D.C. Results show that our Twitter sampling techniques are superior to hashtag sampling alone. They also approximate the ideology and gender distributions derived from on-the-ground surveys, albeit with some bias, but fail to retrieve accurate age group estimates. We conclude that online samples are yet unable to provide reliable representative samples of offline protest.

Credit card transactions and social media posts, repurposed as research data.

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- With **found** data, you have to **reconstruct** it, from the purpose of whoever built the system
- So: who made this, and what were they trying to do?

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Salganik's ten characteristics

Why we want them

Big

Always-on

Non-reactive

Why designs break

Incomplete

Inaccessible

Non-representative

Drifting

Algorithmically confounded

Dirty

Sensitive

Salganik, *Bit by Bit*, ch. 2. Free online.

Big, always-on, non-reactive

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- **Non-reactive:** people are not answering your questions, so they are not performing for you
 - no social desirability bias. But no informed consent either

Non-representative

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- Who *posts* is not who is on the platform

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- So: is a non-representative sample useless?

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Wang, Rothschild, Goel & Gelman (2015), *Int. J. Forecasting* 31(3).

Drifting

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- **The platform changes**

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A measure can stop meaning what it meant,
without anything looking wrong .

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Lazer, Kennedy, King & Vespignani (2014), *Science* 343(6176). This week's paper.

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 - people searched differently because the *product* changed

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Algorithmically confounded

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- The platform is in your causal diagram (week 4)

Inaccessible, and the post-API age

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Freelon (2018), Computational Research in the Post-API Age, *Political Communication* 35(4).

Inaccessible, and the post-API age

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- A design that depends on access you do not control is a design that can **stop working**
- And nobody can check your results afterwards

Freelon (2018), Computational Research in the Post-API Age, *Political Communication* 35(4).

Dirty, and sensitive

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 - the cleaning decisions are design decisions, and they are rarely reported
- **Sensitive:** much of it is about identifiable people who never agreed to anything
 - “it’s public” is not the same as “they consented”
 - aggregation and linkage can re-identify

(week 5)

Incomplete

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Incomplete

- The variable you need is almost never the variable that was recorded
 - platforms record what is useful to them, not what is useful to you
- No demographics. No income. No control variables
- You can sometimes **infer** what is missing. That is a model, and models can be wrong

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A practical interlude

You have a question. Now what?

A practical interlude

You have a question. Now what?

- Most projects in this programme use data somebody has already published
- Knowing what exists, and what it looks like when it arrives, is a real skill

Places things actually live

	Total greenhouse gas emissions		Methane emissions			
	Increased metric tons of carbon dioxide equivalent	% change	Increased metric tons of carbon dioxide equivalent	% change	% of total	% of total
	2012	1990-2012	2012	1990-2012	2008	2008
Poland	67,042.7	-37.1	11,794	-24.8	12.4	10.8
Russia/Fuse	43,910.4	201.4	14,087	106.8	19.9	17.4
Romania	6,254.4	-129.8	2,719	86.4	10.7	22.8
Cabo Verde	—	—	191	62.2	3.0	39.4
Costa Rica	127,389.6	652.4	35,915	137.8	3.7	63.2
Cameroon	180,822.1	3.5	16,516	15.8	13.3	62.3
Canada	1,037,063.9	59.7	156,647	36.5	42.1	26.8
Cayman Islands	—	—	20	52.3	3.2	6.4
Central African Republic	515,134.1	252.3	85,677	102.8	0.5	76.0
Chad	109,795.0	498.7	15,364	65.0	11.5	62.9
Channel Islands	—	—	—	—	—	—
Chile	120,687.9	128.9	19,381	53.3	23.7	43.3
China	12,484,710.6	218.9	1,782,290	72.3	48.0	37.6

Mean and median income by age and sex - EU-SILC and ECHP surveys (online data code: ILC_DIO3\$DV, Source of data: Eurostat)

Table Line Bar Map

IT	TIME	2009 \$	2010 \$	2011 \$	2012 \$
GEO	\$				
European Union (EU6-1958, EU9-1973, EU10-1981, EU12-1992)		14 775 (x)	14 841 (x)	14 968 (x)	15 456 (x)
European Union - 27 countries (from 2020)		:	14 521 (x)	14 632 (x)	14 924 (x)
European Union - 28 countries (2013-2020)		:	14 841	14 968	15 456
European Union - 27 countries (2007-2013)		14 775	14 919	15 841	15 543
European Union - 15 countries (1995-2004)		:	:	:	:
Euro area (EA11-1999, EA12-2001, EA13-2007, EA15-2008)		17 188 (x)	17 381 (x)	17 222 (x)	17 481 (x)
Euro area - 19 countries (from 2015)		16 868	17 047	17 018	17 283
Euro area - 18 countries (2014)		16 986	17 172	17 140	17 481
Belgium		19 313	19 464	20 088	20 280
Bulgaria		2 828	3 017	2 914	2 850
Czechia		7 295	7 858	7 451	7 791
Denmark		25 829	25 868	26 944 (x)	27 184
Germany (until 1990 former territory of the FRG)		18 586	18 797	19 843	19 595
Estonia		6 289	5 727	5 598	5 987
Ireland		22 445	28 512	19 726	19 648
Greece		11 496	11 963	10 985	9 513
Spain		14 795	14 685	13 929	13 868

Also: INE, Eurobarometer, ESS, V-Dem, UCDP, national statistical offices, municipal open data portals, replication archives (Dataverse, OSF).

What a codebook gives you

Question Number: Q52G

Question: How much do you trust each of the following, or haven't you heard enough about them to say:
Opposition Political Parties?

Variable Label: Q52g. Trust opposition political parties

Values: 0-3, 9, 98, -1

Value Labels: 0=Not at all, 1=Just a little, 2=Somewhat, 3=A lot, 9=Don't know/Haven't heard enough,
98=Refused to answer, -1=Missing

Source: Adapted from Zambia 96

*Not asked in BFO,SWZ

Question Number: Q52H

Question: How much do you trust each of the following, or haven't you heard enough about them to say:
The Police?

Variable Label: Q52h. Trust police

Values: 0-3, 9, 98, -1

Value Labels: 0=Not at all, 1=Just a little, 2=Somewhat, 3=A lot, 9=Don't know/Haven't heard enough,
98=Refused to answer, -1=Missing

Source: Zambia 96

Question Number: Q52I

Question: How much do you trust each of the following, or haven't you heard enough about them to say:
The army?

Variable Label: Q52i. Trust army

Values: 0-3, 9, 98, -1

Value Labels: 0=Not at all, 1=Just a little, 2=Somewhat, 3=A lot, 9=Don't know/Haven't heard enough,
98=Refused to answer, -1=Missing

Source: Afrobarometer Round 5

And what it looks like when it arrives

muni_code	municipio	PP2016_06	PSOE2016_06	part2016_06	PP2019_11	PSOE2019_11	part2019_11
40099	Honrubia de la Cuesta	0,595	0,262	0,846	0,476	0,31	0,815
50252	Tauste	0,461	0,26	0,709	0,316	0,313	0,695
25087	Esterri de Cardos	0	0,116	0,717	0	0,133	0,833
40033	Brieva	0,373	0,176	0,879	0,212	0,192	0,786
26098	Munilla	0,438	0,137	0,83	0,349	0,27	0,685
28021	Berrueco, El	0,371	0,121	0,668	0,264	0,168	0,639
37059	Buenamadre	0,516	0,29	0,758	0,305	0,4	0,812
50203	Pastriz	0,292	0,241	0,719	0,173	0,284	0,767
06901	Valdelacalzada	0,389	0,375	0,736	0,224	0,405	0,729
04031	Cantoria	0,334	0,499	0,758	0,213	0,537	0,737
17199	Torroella de Montgri	0,089	0,103	0,605	0,041	0,116	0,704
07027	Inca	0,362	0,23	0,631	0,203	0,263	0,595
08240	Sant Sadurni d'Anoia	0,075	0,158	0,697	0,04	0,18	0,776
19256	Sienes	0,639	0,167	0,837	0,556	0,278	0,783
40157	Pelayos del Arroyo	0,477	0,227	0,936	0,333	0,3	0,769
28104	Paracuellos de Jarama	0,381	0,136	0,794	0,236	0,207	0,795
31195	Orbaizeta	0,211	0,062	0,714		0,065	0,639
06135	Trujillanos	0,508	0,313	0,775	0,265	0,343	0,781
08259	Santa Maria de Palautor	0,09	0,122	0,695	0,041	0,145	0,768
12095	Ribesalbes	0,349	0,296	0,781	0,275	0,361	0,761

Spanish municipal election results. What is the unit? What is missing?

Before you commit to a data source

- Who made it, and **why**?

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Six questions. They are also, more or less, Memo 2.

Roadmap

From topic to research question

Theory, mechanisms, levels

Where does the data come from?

Ten things about big data sources

So where do you actually get data?

Paper: The Parable of Google Flu

Before next week

Why this paper

- Three pages, no equations
- Measurement validity, drift, and algorithmic confounding **in one case**
- And a well-resourced team getting it wrong in public

In pairs

1. What was GFT actually **measuring**?
2. Whose behaviour changed: the public's, or Google's?
3. Would **more data** have fixed it?
4. What would you have had to do in 2009 to see this coming?

Groups of four: redesign it

You have Google's search logs and CDC data, in 2009.

Build something that does not break the same way.

Seven minutes. One change, and what it buys you.

The design lesson

Your instrument was built by someone else,
for their own purposes,
and they will keep changing it.

Which is a measurement problem. That is next week.

Roadmap

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Before next week

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- **Read:** Müller-Crepon, Hunziker & Cederman, *Roads to Rule, Roads to Rebel*
 - Longer than Google Flu. Skip the statistics, read the measurement
- **Memo 2**, in pairs, Aula Global by **Monday night**:
 - Take a source of found data you could realistically get hold of
 - Who produced it, and why? Who or what is missing from it?
 - Name one question it could answer well, and one it could not
 - Half a page

Questions?